Reflections on MathematicalMonk

Lessons learned from teaching with YouTube

Jeff Miller

CORES Symposium | Stanford, CA | May 20, 2025

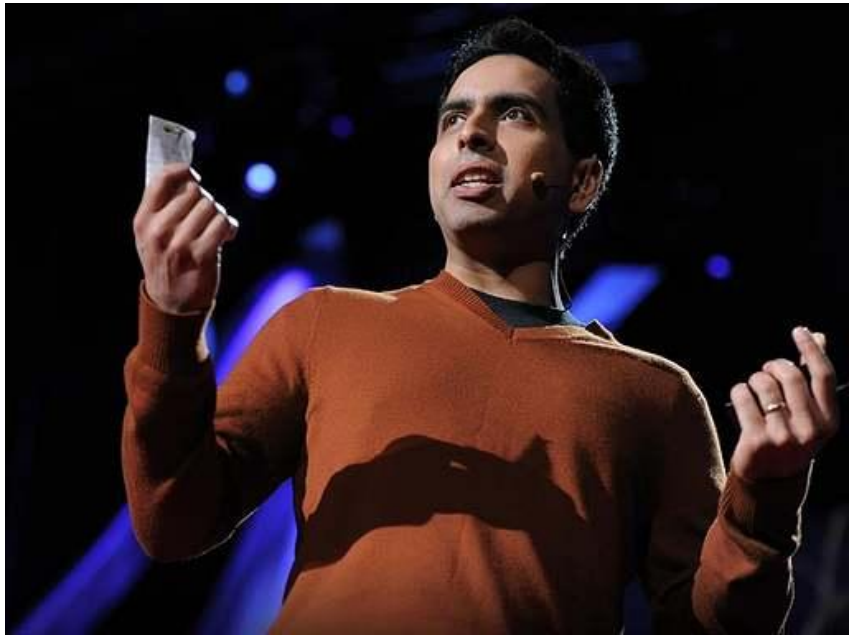
How it started…

- In 2011, I was a PhD student at Brown University.
- Had the opportunity to teach machine learning in summer 2011 and information theory in fall 2011.



How it started…

- I was inspired by a TED talk given by Sal Khan on Khan Academy.
- Had a great experience with MIT OpenCourseWare – particularly, Gilbert Strang's video lectures on linear algebra.



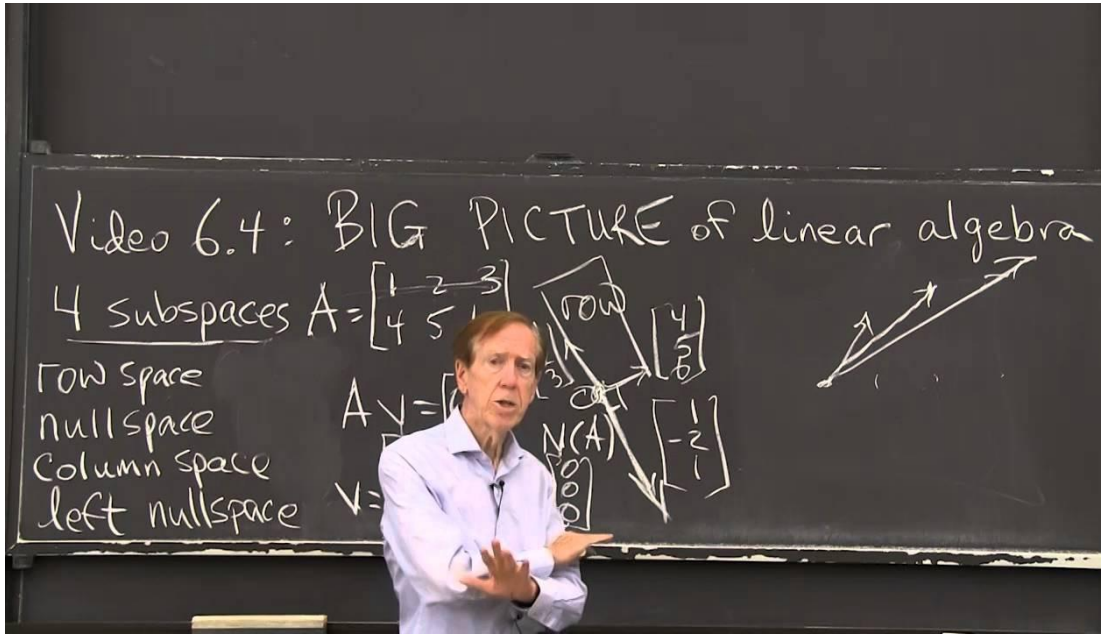
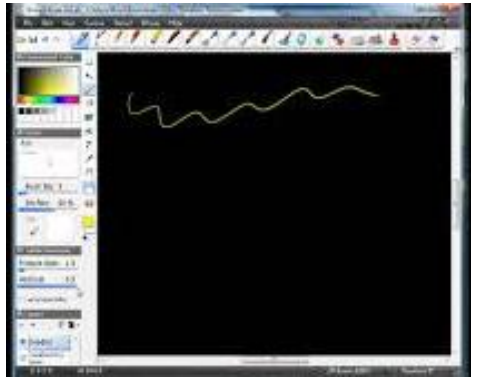


Image credits: TED.com, MIT OpenCourseWare (ocw.mit.edu)

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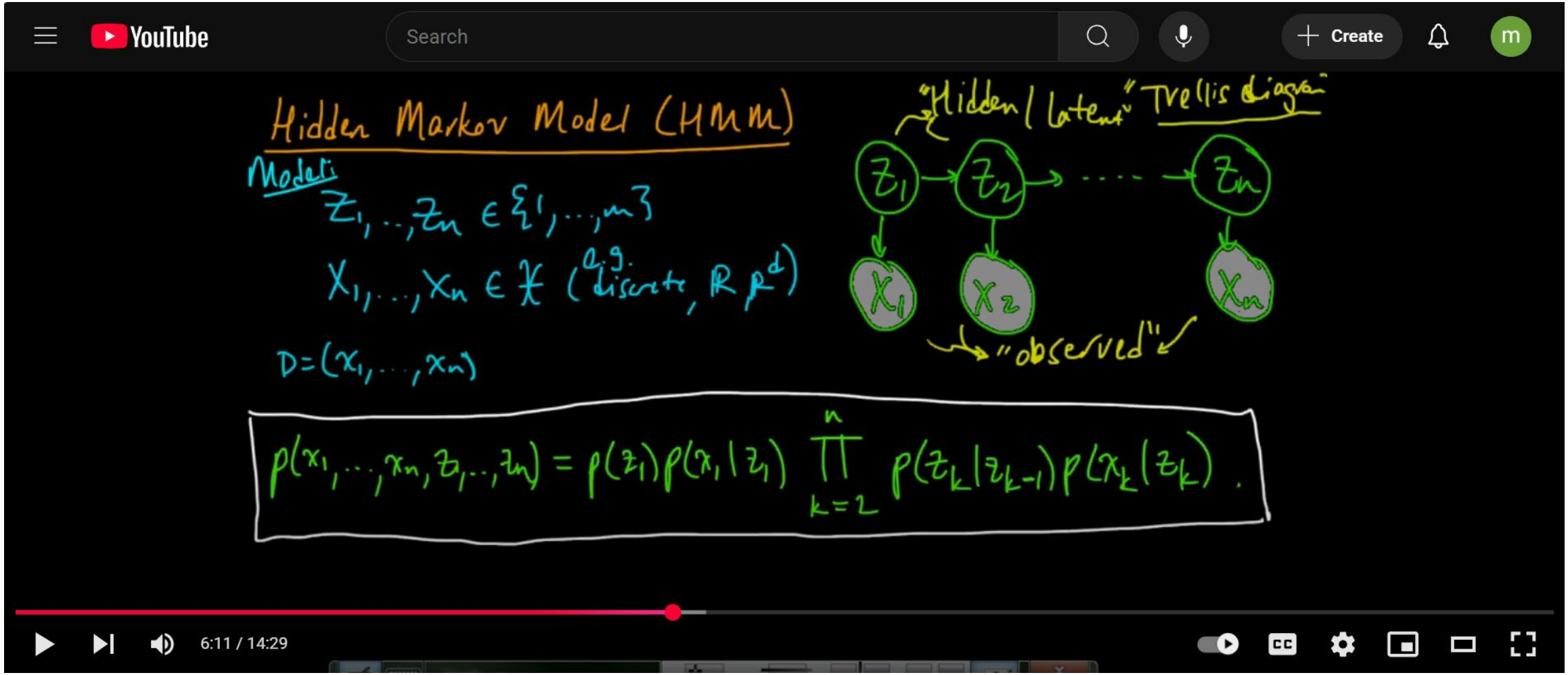
- Loved the idea of “flipping the classroom”.
- Decided to create video lectures for my classes and put them out there for anyone to watch, available online for perpetuity.
- Goal was to create Khan Academy style videos on more advanced topics.



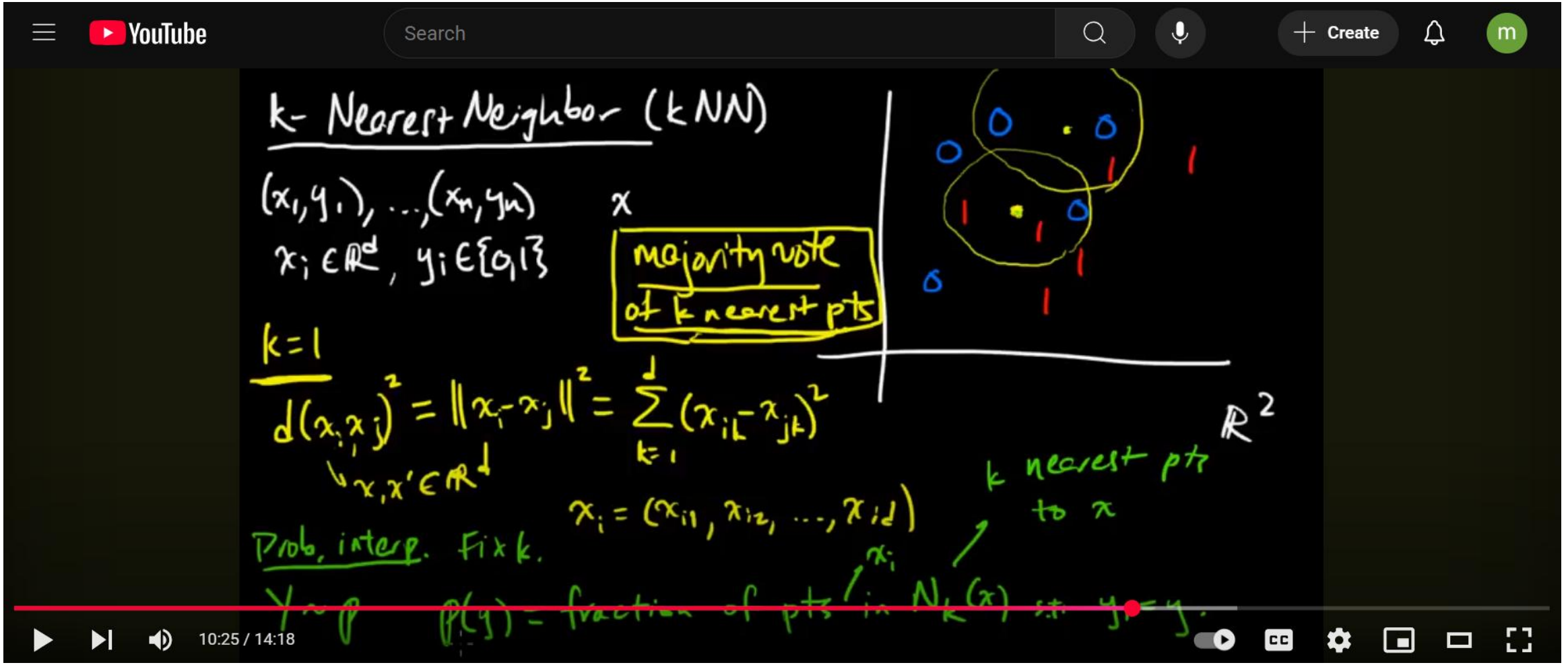




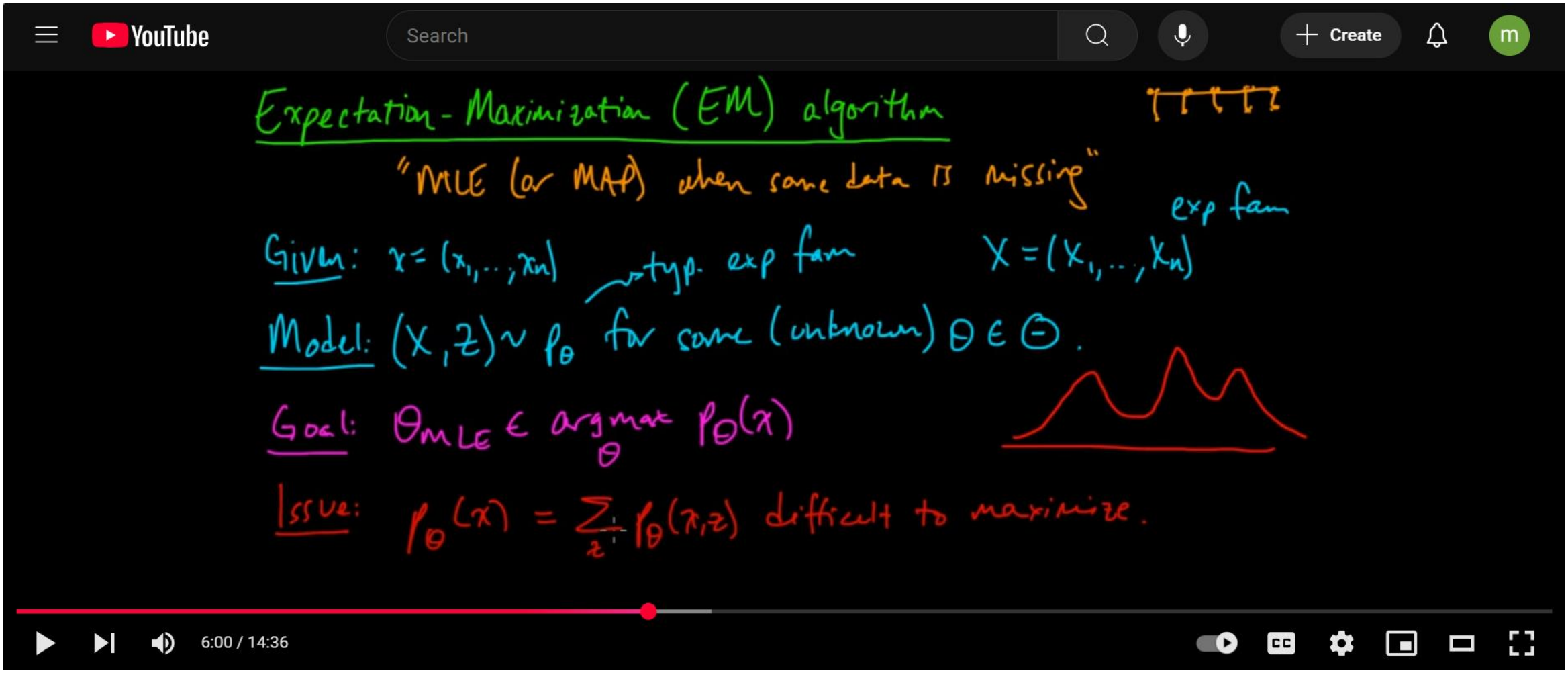
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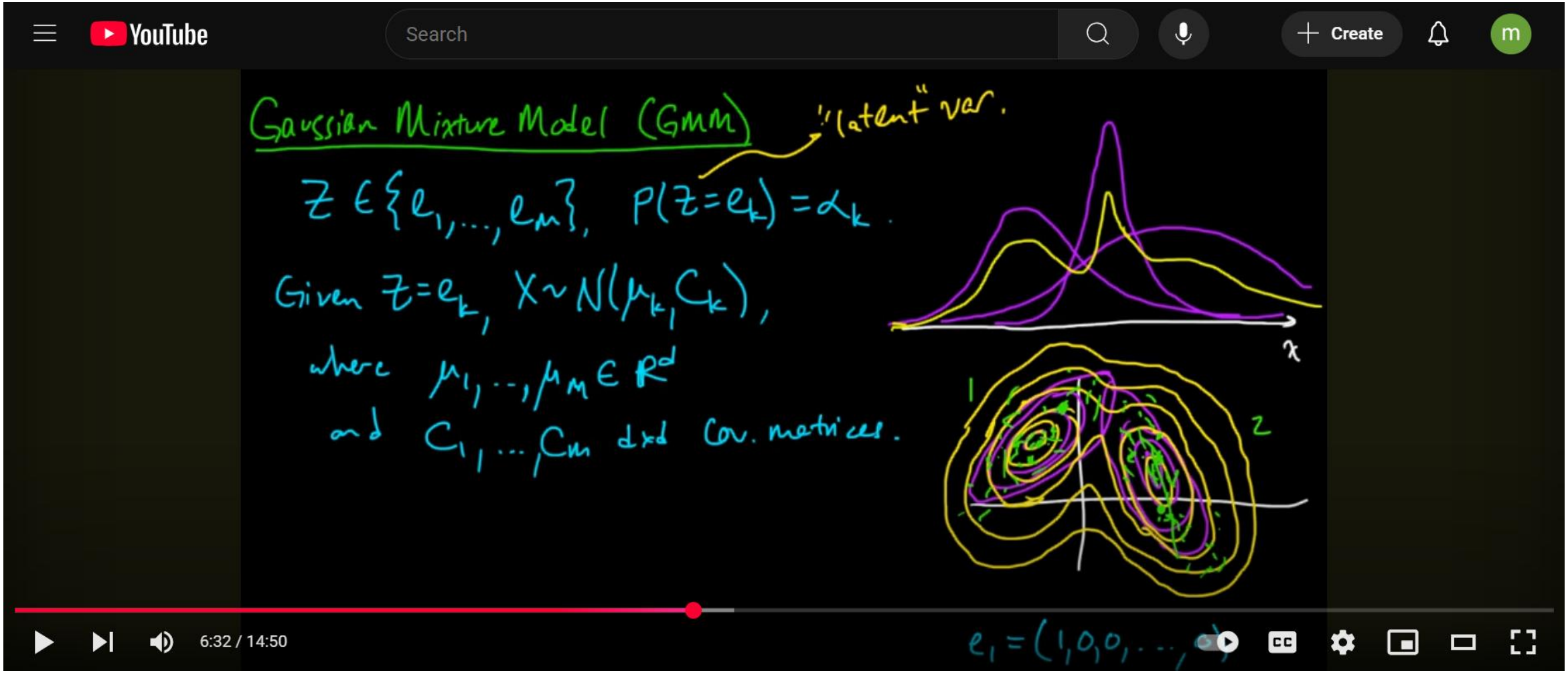
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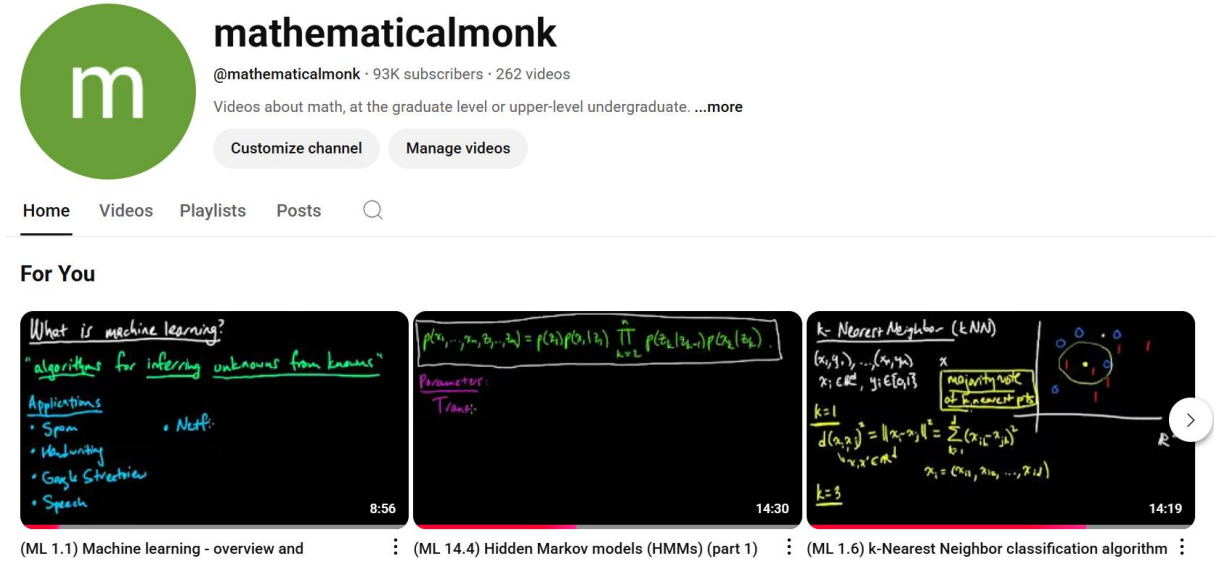


… lots of videos



If you build it, they will come

- Recorded 262 videos over 7 months (Apr 2011 – Oct 2011).
- Created the mathematicalmonk channel on YouTube.
- Today there are > 93,000 subscribers.
- Over 10,000,000 views to date (around 60K views/month on average).



Generally positive response from viewers

Most recent three comments:

- *“it helped me finaly understand this method!!! thank you so much”*
- *“We of the present really do stand on the achievements of the past. Thank you man”*
- *“Please come back...”*



A range of feedback as well

- *“Thanks for the video, but in fact you don't need to factor both C and D, just factoring one of them is enough! As Schur did in his paper (1911).”*



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Pay no attention to the man behind the computer

“I really would like to know, who is this guy? Such brilliant videos and no idea about the guy behind these videos...”



Image credit: Fahim Hamim, Vecteezy.com

Pay no attention to the man behind the computer

Why did I make MathematicalMonk anonymously?

1. Fear: If people hated it or I did something stupid, then I wouldn't be embarrassed.

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Later, it kind of became fun to keep it anonymous. I added a link to the channel from my webpage, so people could find me if they really wanted to.

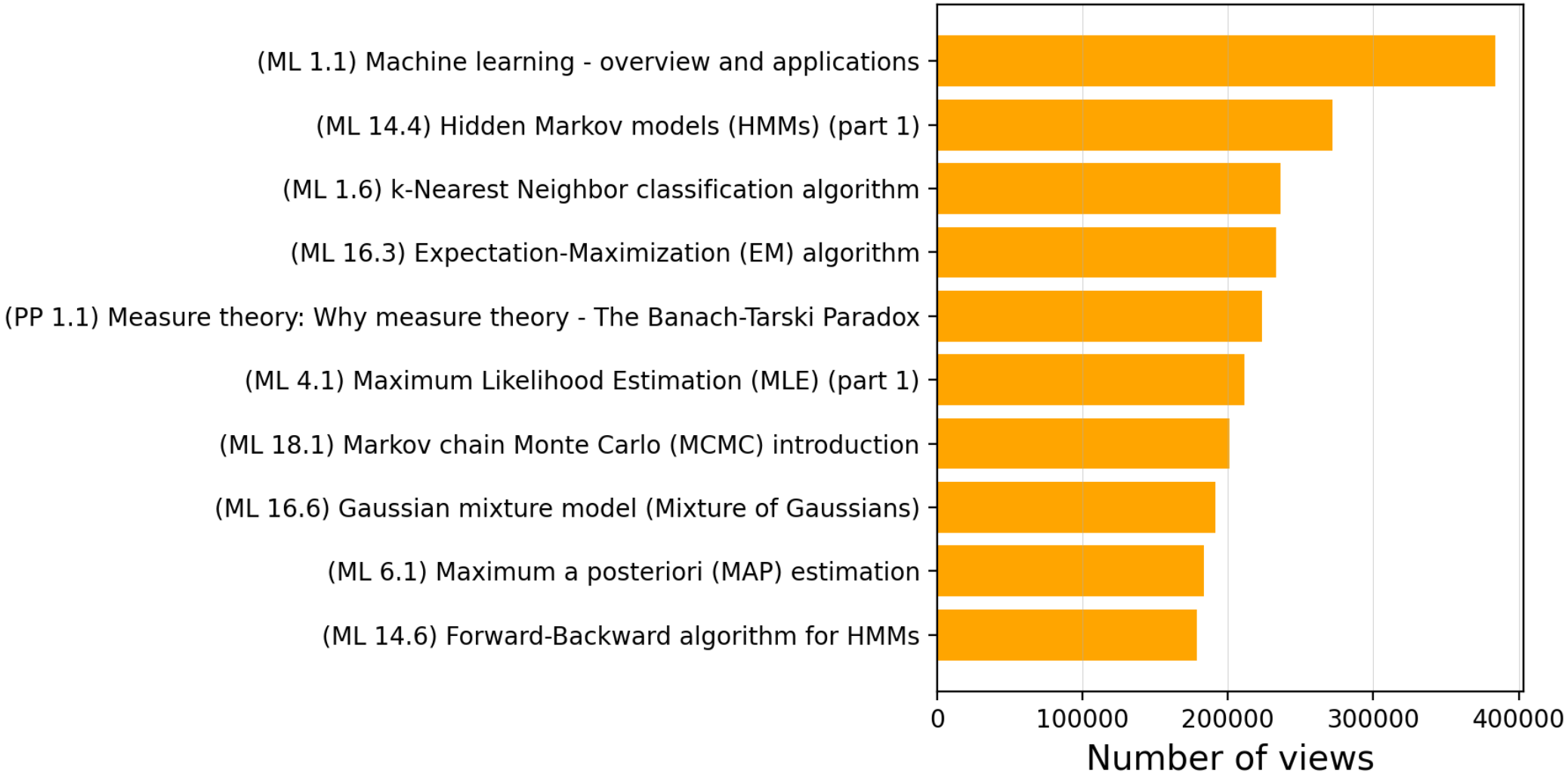
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Channel stats and analytics

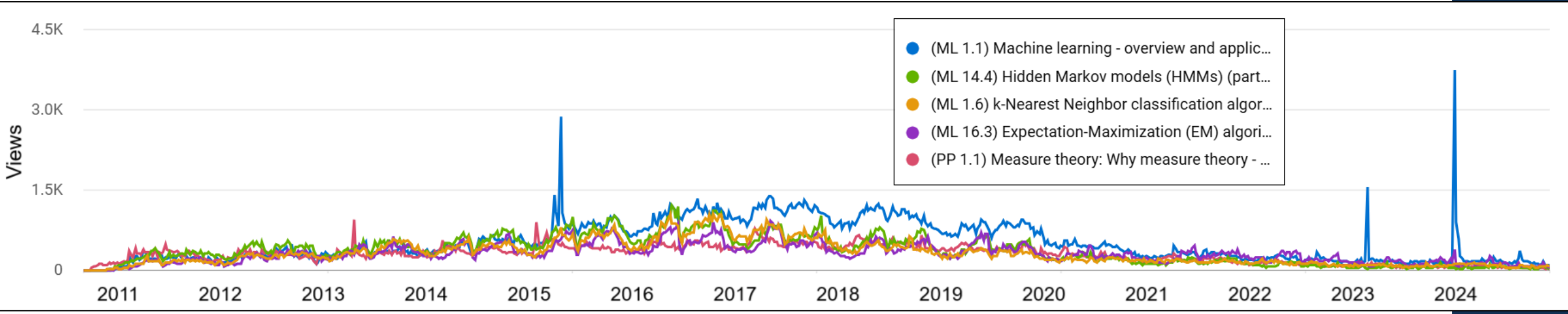
What are people watching?

Most watched videos on the channel



Number of views over time (weekly totals)

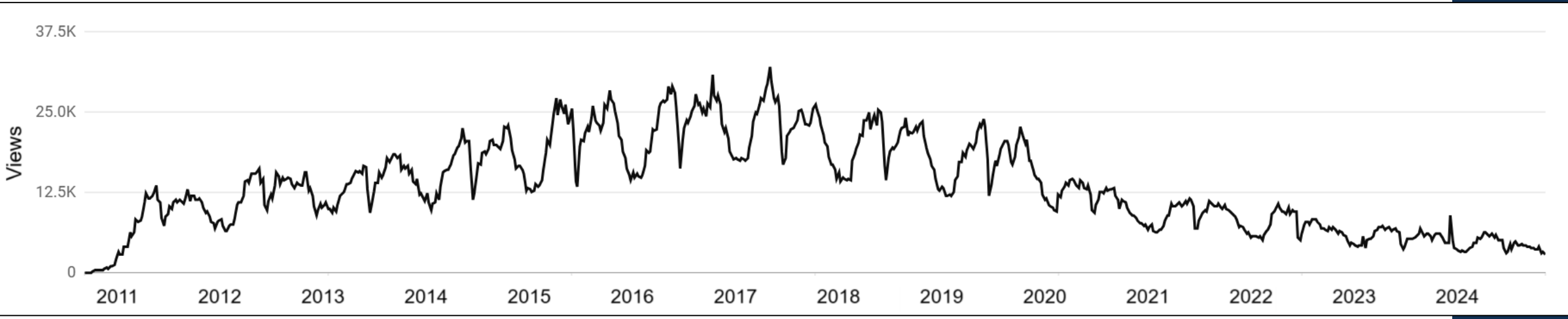
Most watched videos on the channel



- ML 1.1 spiked on three specific dates:
  - Oct 28, 2015: 1738 views
  - Aug 7, 2023: 963 views
  - June 12, 2024: 2629 views
- I have no idea what happened on these dates... if anyone knows how to find out, I'd be curious to know.

Number of views over time (weekly totals)

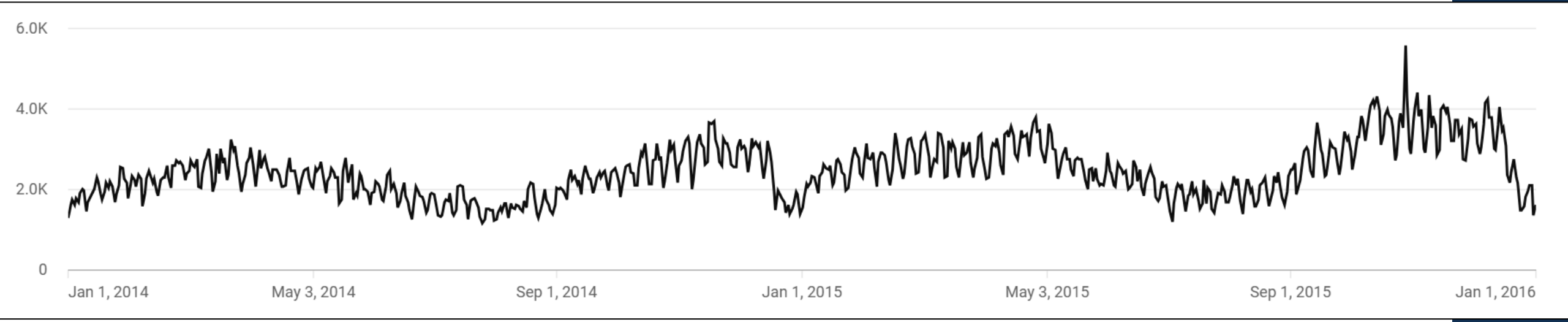
Total over all videos on the channel



- There are dips in view counts during summer and winter break.
- The cyclical trend correlates with school semesters (spring, fall).
- Views have trended downward since 2017, perhaps due to increasing interest in deep learning rather than machine learning, statistics, and information theory.

Number of views over time (daily totals)

Total over all videos on the channel (2014-2016)

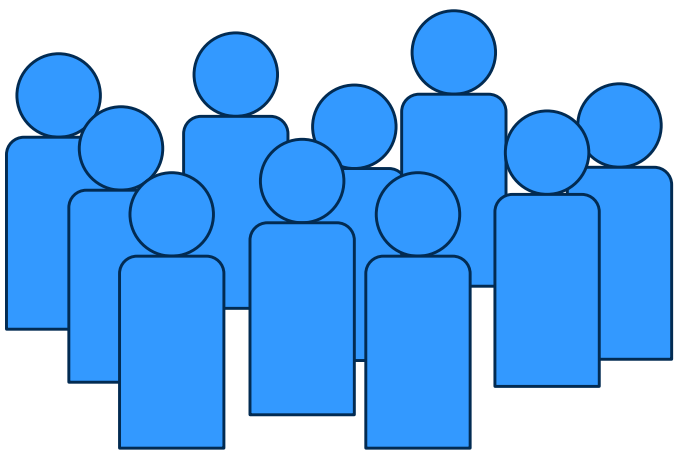


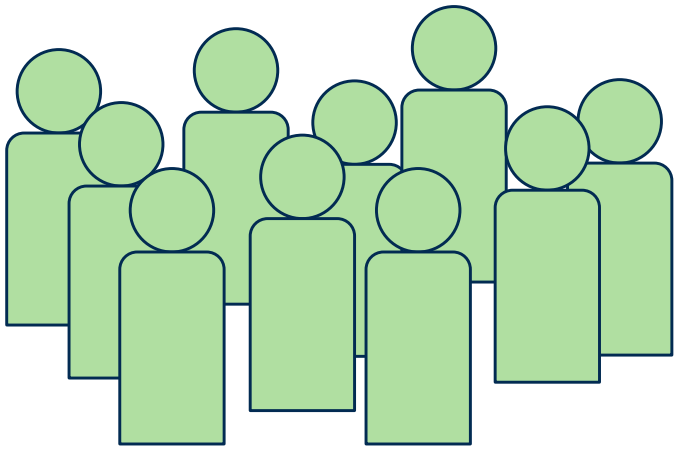
- Zooming in to the daily totals, we see another cyclical trend.
- This cycle correlates with week.
- Specifically, there is a dip in view count during Friday-Saturday.

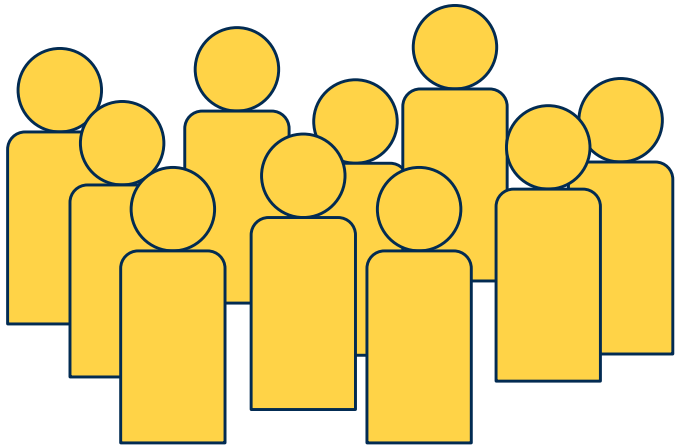
Who is watching?

From what I can tell:

- Students studying a particular topic for a class they are taking
- Self learners who go through many videos
- Casual viewers who watch one or two videos







Reflections on MathematicalMonk

Lessons learned from teaching with YouTube

Lessons learned:

1. Flipping the classroom doesn’t work

The allure of the flipped classroom

Flipping the classroom makes a lot of sense:

- Lectures are largely a one-way flow of information.
- So, why not have students watch videos outside of class, and use classroom time for more interactive activities?
  - Students can absorb the material from videos at their own pace!
  - Valuable in-person class time can be used for discussion and exercises!

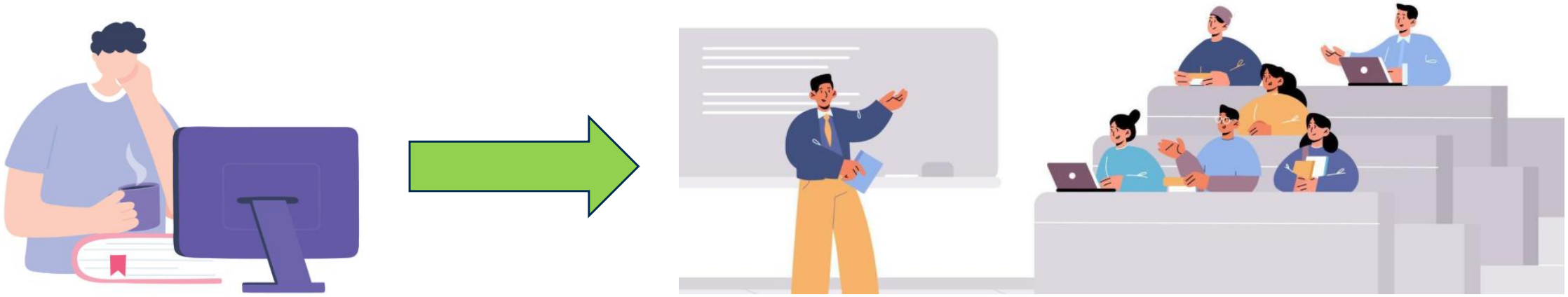


Image credits: Giuseppe Ramos, Yuliya Pauliukevich, Vecteezy.com

So, what's the problem?

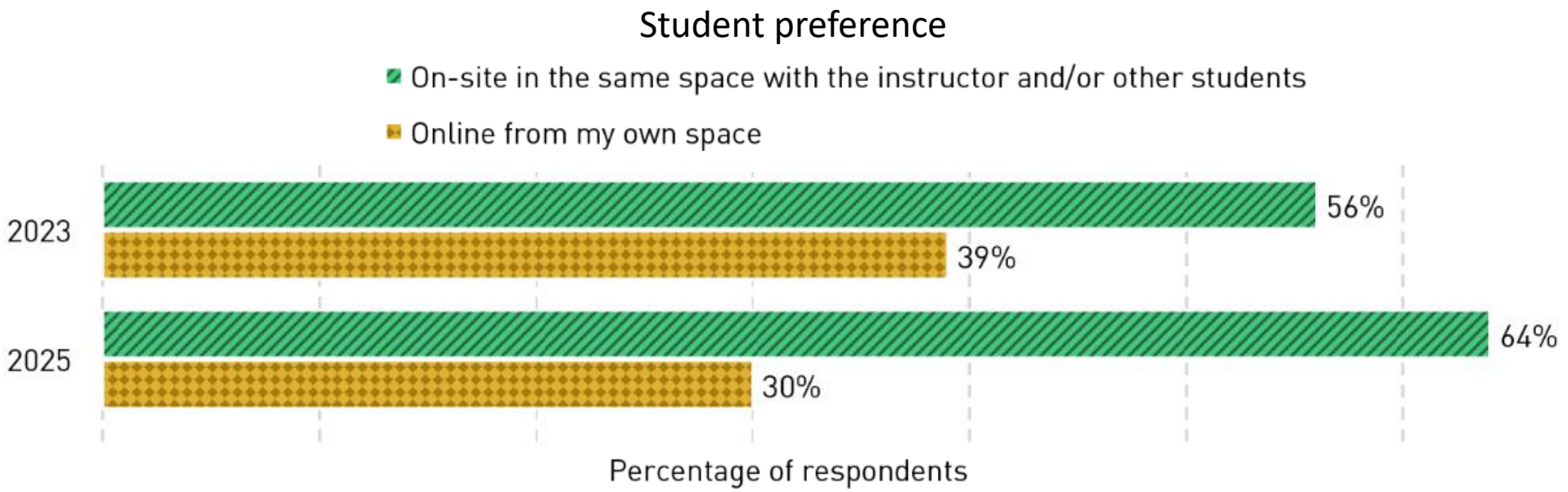
- Unfortunately, flipping doesn't work for most students, in my experience.
- Perhaps 20% of students would thrive in a flipped classroom.
- But most students don't consistently watch the videos ahead of class.
- Then, they are lost and just disengage during the in-class activities.



Image credits: Md Sujan Mia, Vecteezy.com

What do students prefer?

Most students also prefer in-person lectures over remote lectures.



Nicole Muscanell and Kristen Gay. *2025 Students and Technology Report: Shaping the Future of Higher Education Through Technology, Flexibility, and Well-Being*. Research report. Boulder, CO: EDUCAUSE, April 2025. https://www.educause.edu/content/2025/students-and-technology-report

My current approach: Something in-between

- Present new material during lecture
- 1-2 in-class group exercises per lecture period
- Lab periods (led by TA) with hands-on exercises
- Record lectures and provide them online



Image credits: Atanu Tanmoy, Vecteezy.com

One weird trick to boost student engagement

- **1-2 in-class group exercises per lecture period**
  - Give students an exercise on lecture material
  - Have them work for ~10 minutes in groups of 2-3
  - Go over it afterwards on the board
- Students LOVE this!!! Very positive response in course evaluations.
  - Provides real-time feedback on whether they are getting it or not.
  - Helps practice for exams and see what kind of problems you create.
- It is especially helpful for teaching via Zoom.



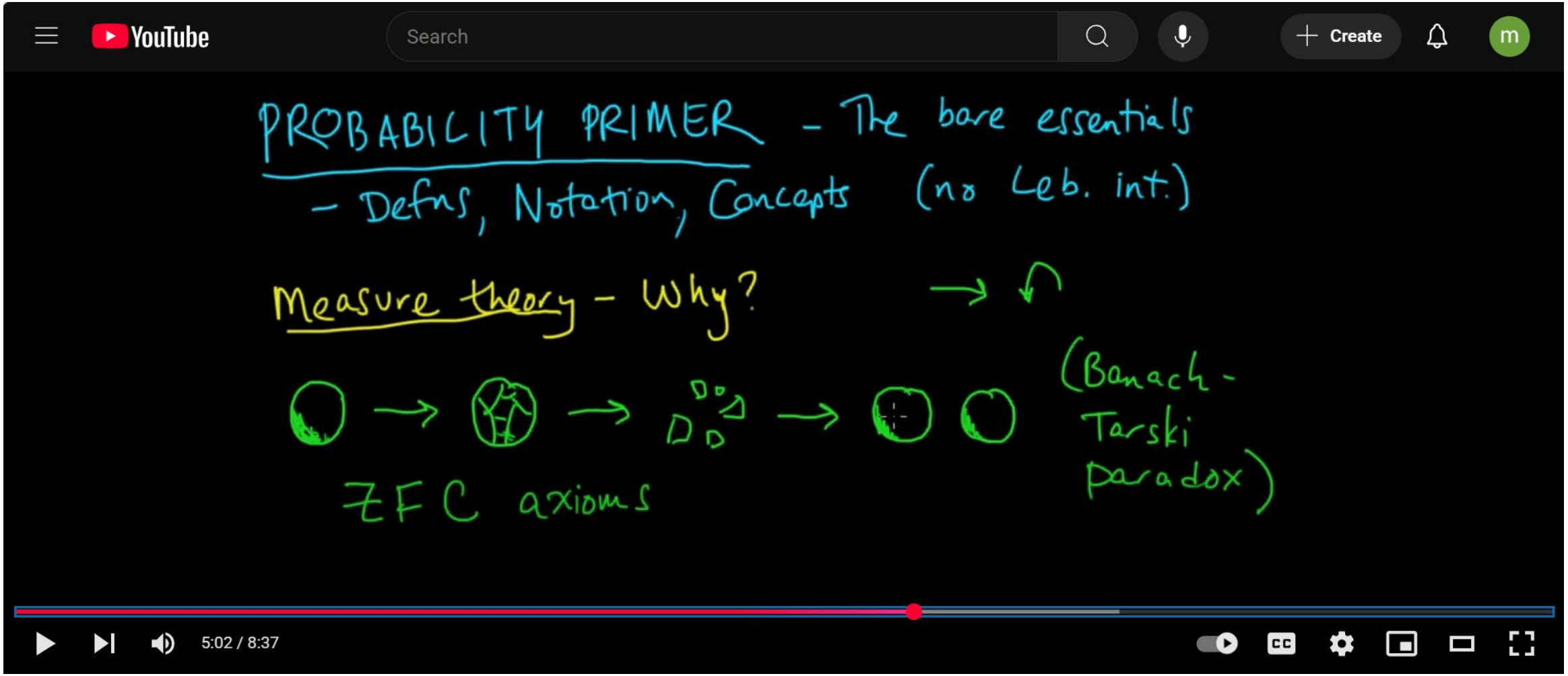
Image credits: Yuliya Pauliukevich, Vecteezy.com

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2. A messy blackboard can be better than a polished presentation

“Blackboard style” of videos



Why use the blackboard style?

- MathematicalMonk videos use the blackboard style of Khan Academy.
- Why is this so effective?
  1. **Slows you down:** It forces you to think through each step at the same pace as the students. Getting stuck at a tricky point is helpful for the students!
  2. **Reveals the thought process:** Students can better understand the logical steps you take and see the little tricks that would get glossed over in slides.
  3. **Shows that you are human:** Math/stats can be very intimidating… Seeing you make mistakes and get stuck is psychologically helpful for students.
- The same rationale applies to in-person teaching on the blackboard.

Why use the blackboard style?

- For video lectures, specifically, there are some other benefits:

4. **Seeing the speaker can be distracting:** A disembodied voice is somehow more personal, almost like it is inside your head. (Sal Khan used to say this.)

5. **It is much easier to create videos:** Requires minimal equipment, and you don’t have to worry about your appearance.

Slides have their place, too…

- I once taught an entire semester of machine learning in-person on the blackboard. **…I do not recommend it!**
- Sometimes, writing things out is too slow. For non-technical content, slides are preferable since students can absorb this material more quickly.
- For classroom teaching, a combination of the two seems optimal to me:
  - Slides for things that don't require in-depth thought.
  - Blackboard for mathematical derivations, etc.





Image credits:
Worapong Saleewong,
Alexis Lis, Vecteezy.com

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3. Impactful content does not need to take a lot of time to create

Low tech can be high quality

Impactful content does not need to take a lot of time to create

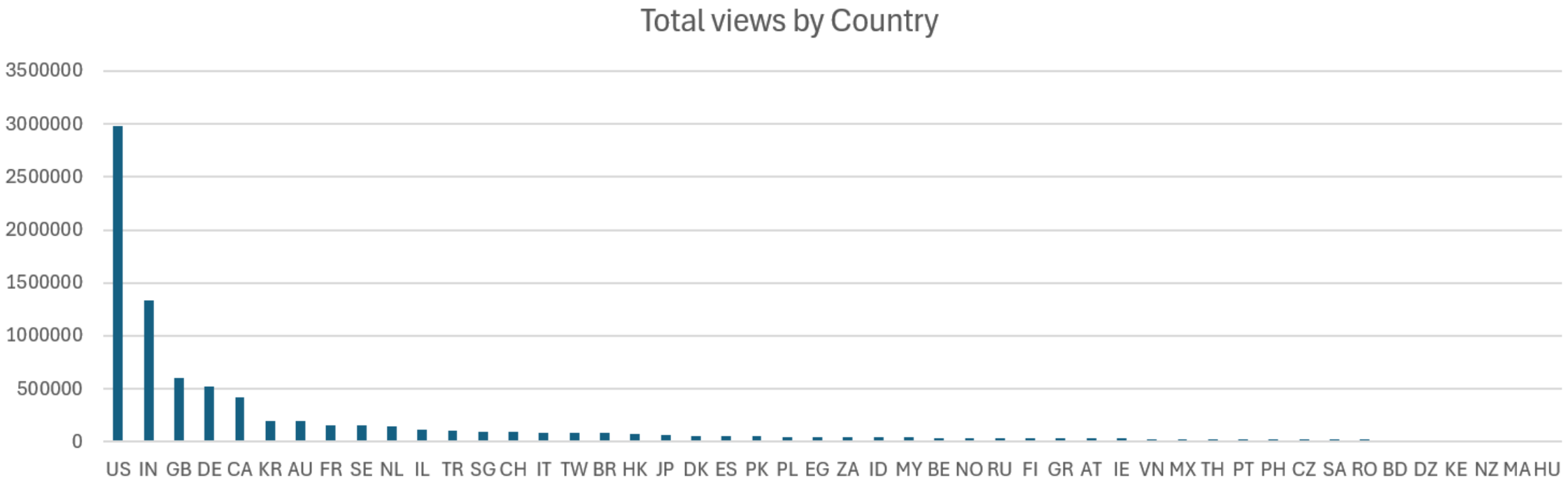
- It took me around 2 days to make each module of 6-7 videos – one day to plan and write out my notes, and one day to record videos.
- I recorded each video in “one take”, but often paused the video while recording to gather my thoughts. Sometimes, I had to redo a video if I didn’t like it.
- But I didn’t edit videos for the simple reason that I didn’t know how and didn’t want to spend the time to learn how.
- In retrospect, this enabled me to create videos much more quickly since I spent zero time editing and polishing them.
- Flashy videos and pretty graphics are nice to look at and may get more views.
- However, simple content-focused videos can be highly effective and impactful.

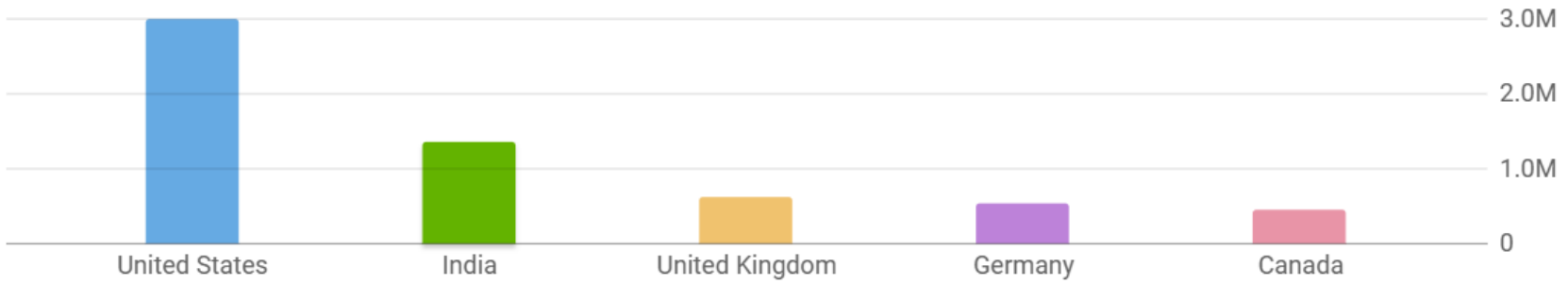
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Vision for open education resources?

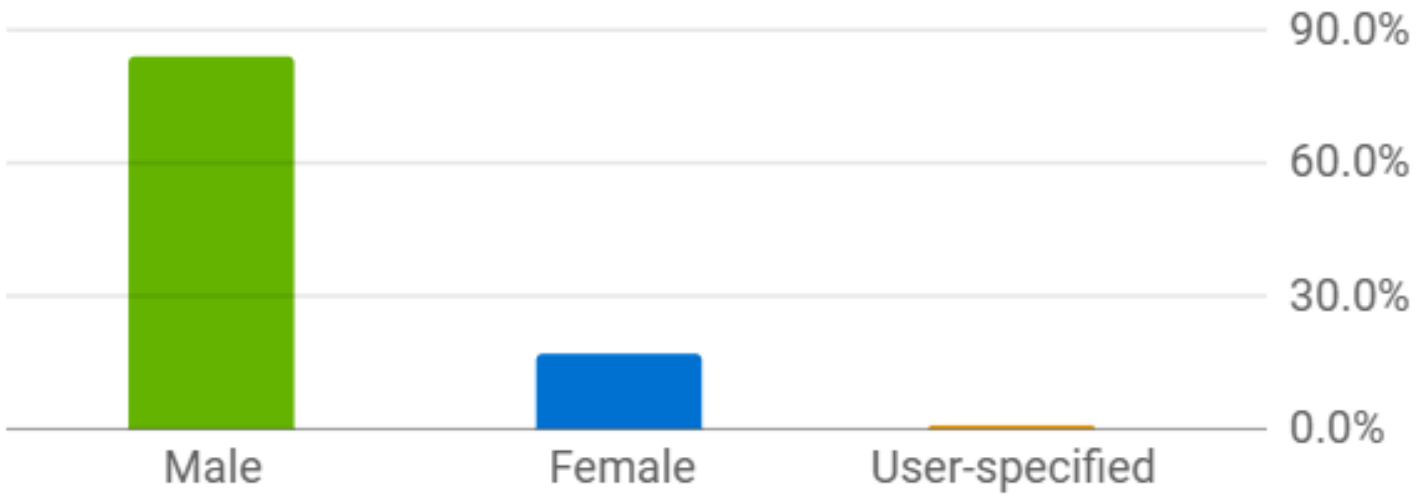
Demographics (views by country)





Demographics (gender)

Total views by gender



Future of open education?

- Potential for global impact that is not yet fully realized?
- Potential for reducing inequities that is not yet fully realized?
- Open resources work well for self learners and for augmenting in-person classes.
  - But for most people, the in-person experience seems essential for motivation & focus.
  - There will remain a need for in-person degree granting programs.
- Are open education resources primarily used by students taking classes on these subjects? If so, what could extend their reach to other populations?

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